

# ADAM TORRES ENCARNACION

University Park, PA | Email  | LinkedIn  | GitHub  | Portfolio 

## EDUCATION

### The Pennsylvania State University

M.S. in Artificial Intelligence

**Expected:** Dec 2027

### The Pennsylvania State University

B.S. in Data Science, B.S. in Statistics

**Expected:** Dec 2026

**GPA:** 3.7

## WORK EXPERIENCE

### Amazon

Software Engineering Intern

May 2026 – Aug 2026

Cupertino, CA

- Designed C++ infrastructure for a dual-arm fiber-cleaning robot targeting \$28.6M/year in savings across 85 data centers
- Built a 19-route Coral robot control plane over SSH that kept the 1 kHz control loop server-side for remote clients
- Centralized 100% of robot geometry and vision labels from a 3-model LLM-as-judge pipeline in DynamoDB and S3
- Bootstrapped a Model Context Protocol (MCP) server on SQLite compiling 74 doc sources into a cited knowledge base
- Hardened CI/CD with 3 pipeline stages, 240+ unit tests, and content-addressed SSH releases, stopping host drift

### IBM

AI Engineering Intern

May 2025 – May 2026

Remote

- Shipped a REST LLM microservice that generated documentation for 3100+ scripts and reduced team costs by \$223,000
- Refined an AI file-chunking pipeline with Llama-4 and GPT-4o-mini to process 7+ large-scale internal requests
- Deployed serverless Azure Function Apps with automatic codebase scaling, cutting the manual review workload by 78%
- Helped triage ~500k criminal records spanning 20+ years with a LangChain chatbot built on Python, cutting review time

## PROJECTS

### [terrari.ai](#) (HackPrinceton '25 xAI 1st place winner) | C#, Grok

- Crafted Terraria NPC agents framework that turns user words into action, earning \$8k against 800+ participants
- Implemented C# registry of 7 typed world actions/tools for agents, powering coordination for up to 15 parallel units

### [simulacra](#) (YHack '26 K2 Think V2 1st place winner) | Python, FastAPI, LangGraph, React

- Designed a multi-agent orchestration engine in FastAPI with LangGraph StateGraph, running concurrent cognitive loops via asyncio with shared memory streams and Pydantic-validated structured outputs
- Built per-agent memory retrieval with temporal decay scoring, threshold-triggered reflection, and Deffuant bounded-confidence opinion dynamics to model echo-chamber formation across 50+ agents streamed over Socket.IO

### [nexdraw](#) (NexHacks '26 ByteDance 2nd place winner) | TypeScript, Next.js, Gemini, PostgreSQL

- Placed 2nd of 1500+ solo with a tldraw canvas that auto-completes sketches in place via Gemini AI image generation
- Classified chat vs draw with Gemini, then cropped the subject with Hugging Face Grounding DINO on CUDA (PyTorch)
- Piped voice prompts through OpenAI Whisper and gated every generation behind an accept/reject overlay on the canvas
- Persisted tldraw snapshots as PostgreSQL JSONB on a 2s debounce, writing PNG previews and skipping saves offline

## LEADERSHIP EXPERIENCE

### Claude Builder Club

Vice President

Aug 2025 – Jan 2026

University Park, PA

- Brought Claude AI to Penn State, drawing in 260+ club signups in its first month (the fastest growth in school history)
- Taught AI tooling (agent harnesses, skills, hooks, MCPs, multiplexers) to 170+ students, with 90% returning for follow-ups

### Nittany AI

Director (AI)

Jan 2025 – Jan 2026

University Park, PA

- Led a research reading group for the largest AI club on campus, with 40+ students per semester and 85% retention
- Covered papers [YOLO](#), [SAM3](#), [Attention Is All You Need](#), and [Adam Optimizer](#) across 5 months of student-led seminars
- Partnered with 3+ AI labs to host PhD panels and professor-led research demos, growing club membership by 247%

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, C, C#, SQL, TypeScript, HTML/CSS, R, Lua

**Frameworks:** PyTorch, OpenCV, Hugging Face, Pandas, NumPy, FastAPI, React, Node.js, Next.js, LangChain/LangGraph

**Developer Tools:** Git, Linux, VS Code, Vim/Neovim, Pi (harness), Cmux (terminal multiplexer), Agile/Scrum

**Databases:** PostgreSQL, MySQL, SQLite, Redis, MongoDB, DynamoDB

**Cloud/Infra:** AWS (Lambda, S3, EC2, Bedrock, DynamoDB), Azure Functions, Vercel, Docker, CI/CD (GitHub Actions), Supabase